

5.6 Cancer Genetics

BE ABLE TO EXPLAIN HALLMARKS OF CANCER

▼ What is cancer?

- Cancer is a condition where **cells in a specific part of the body grow and reproduce uncontrollably**.
- The **cancerous cells can invade and destroy surrounding healthy tissue**, including organs.

▼ What is tumour heterogeneity?

- Tumour heterogeneity describes how **differences can exist both within tumours and between different types of tumour**.

▼ What is the difference between intratumour and intertumour heterogeneity?

- **Intratumour heterogeneity** describes how **different groups of cells called "sub-clones" can come together and constitute a single tumour**
- **Intertumour heterogeneity** describes how **no two people have the same cancer**

▼ What is the difference between a driver and a passenger mutation?

- **Driver mutations** are those that **drive cancer initiation and progression**
- **Passenger mutations** are those that **do not drive cancer initiation and progression**

▼ What is a cancer risk gene?

- It is a gene that is **at high risk of suffering from a driver mutation**.
- These genes are normally protective against cancer, but once they mutate, we become at risk of cancer
- There are approximately **743 cancer risk genes**

▼ What is a proto-oncogene?

- A **normal, benign gene** that **codes for a protein that promotes cell growth and proliferation**
- When there is a **driver mutation in a proto-oncogene**, it becomes an **oncogene**

▼ What causes a proto-oncogene to become an oncogene?

- A **driver mutation**.
- i.e a driver mutation can occur in the **KRAS gene** which causes it to **continually proliferate and lead to cancer**.

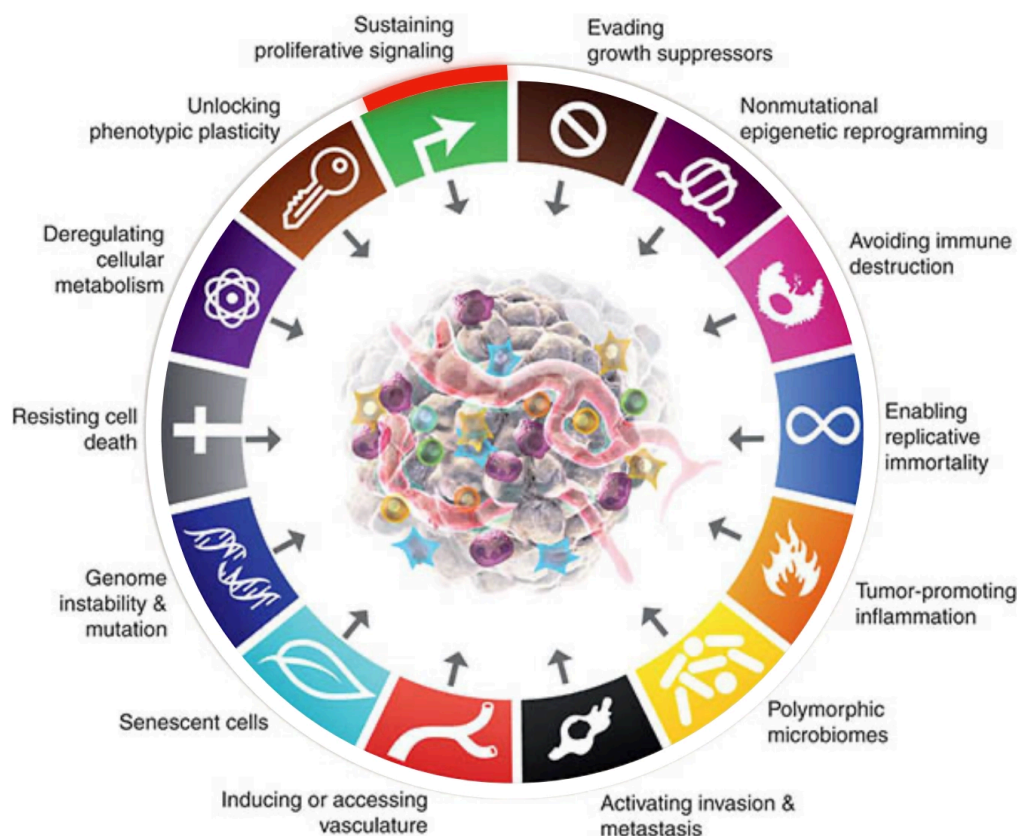
▼ What is a tumour suppressor gene?

- A type of **normal gene** that **produces a protein that helps to limit cell growth and proliferation**
- **p53** is an example of a **tumour suppressor gene**, as it makes the **tumour protein** which suppresses proliferation
 - p53 → p21 protein → binds to CDK-cyclin complex cell proliferation is stopped

▼ What is the Knudson hypothesis?

- The hypothesis that states that **both alleles** for tumour suppressor genes need to be **inactivated to cause phenotypic changes**.
- It is also known as the **two-hit hypothesis**

▼ Image showing the hallmarks of cancer.



▼ What is a germline mutation?

- Mutations that **occur in the germ cells (egg or sperm)**
- They are **passed on from parents to offspring**

▼ What is a somatic mutation?

- Mutation in any cells that **aren't in the germ cells** (not egg or sperm)
- They are **not passed on from parents to offspring**

▼ Why is it difficult to treat cancers?

- The difficulty arises from a **problem with selectively targeting cancer cells, without targeting normal host cells.**

▼ How can this be worked around (relatively)?

- Chemicals can be engineered to kill fast dividing cells
- The problem is, some healthy cells (i.e hair follicles, GI tract cells) are also fast dividing
- This is why **chemotherapy** can lead to **side effects such as vomiting, hair loss** etc.

▼ Recall three treatments for cancer.

- **Surgery**
- **Chemotherapy**
- **Radiotherapy**
- **Targeted therapy**
- **Immunotherapy**

▼ How is Whole Genome Sequencing used in genetic screening?

- It can be used in **pre-malignant monitoring** to **detect when a person is at risk of cancer** (i.e due to family history)
- It can be used to **diagnose and classify cancer** in those who already have it
- It can be used in **pharmacogenomics** to **determine the best possible treatment for patients who already have cancer**

▼ What is the Cancer Gene Census (CGC)?

- This is an ongoing effort to **catalogue all genes that have been causally linked to cancer**

▼ **NOTE: Different types of cancer genes require different number of driver mutations to lead to cancers.**

